Abstract 4317

A Model-Agnostic AI Agent for Automated Molecular Tumor Board Reports: A Pan-Cancer Proof-of-Concept

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Background

Deterministic AMP/ASCO/CAP tiering (Li 2017) takes the higher tier between OncoKB and CiVIC, conflating knowledge-base (KB) agreement with one-sided coverage or with high-tier CiVIC evidence in a different tumor type, risking false Tier I calls. Recent large language model (LLM) variant-tiering benchmarks report over-classification, motivating an architecture restricting LLM use to KB-discordant variants only.

Methods

We built a targets-orchestrated R pipeline integrating VEP, OncoKB, CiVIC, ESCAT, and AMP/ASCO/CAP tiering. Concordant variants resolve deterministically; KB-discordant variants escalate to a model-agnostic agent emitting a unified tier, confidence, and rationale (JSON-schema validated). We benchmarked it against the deterministic baseline on 112 real variant-tumor cases (99 COSMIC plus 13 cross-tumor stress), measured within-model (3 runs) and cross-model reproducibility (Opus 4.7, Sonnet 4.6, Haiku 4.5), and prospectively deployed it on 62 consecutive late-stage cancer patients (8 monthly TruSight Oncology 500 batches, 11 cancer types).

Results

Output structure and reproducibility are summarised in the Table. Disagreements concentrated at documented rule blind spots: when CiVIC carried different-tumor evidence the agent correctly down-tiered RET M918T NSCLC from Tier IA to IIC; the dominant disagreement (20 of 112) was the Tier III VUS to Tier IID boundary for oncogenic variants lacking therapeutic level. All 62 deployed cases completed end-to-end with auditable per-case reconciliation logs.

Metric	Value
Schema-valid JSON output	112 / 112
Agent-baseline tier agreement	90 / 112 (80.4%)
OncoKB-only Tier I/II override	0 / 70
Within-model reproducibility (Fleiss kappa)	1.000
Cross-model reproducibility (Opus / Sonnet / Haiku)	0.717

Conclusions

Confining LLM invocation to KB-discordant variants matches deterministic reproducibility on safety-critical cases while detecting tumor-specificity errors the rule cannot encode, supporting deployment as second-opinion decision support in molecular tumor boards. Code, prompts, and schemas are open source.

Clinical trial identification

Editorial acknowledgement

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